Review

Wednesday, August 26, 2026 9:18 AM

p4 Ass2

Closed subset of compact set is compact

WTS: $K \subseteq \mathbb{R}^n$ compact

$F \subseteq K$ closed

To show F compact

By Heine Borel, K closed & bounded

F is subset of K,

therefore F bounded

F closed by assumption,

thus F compact

Can also use convergent seq

Ass3 p2

$\mathbb{R}_4[x] = \text{span}(1, x, x^2, x^3, x^4)$

$D : V \to V$

$p \to p'$

$\ker D = \text{span}\{1\}$

$\text{im } D = \mathbb{R}_3[x]$

why not
span $\{[1,0,0,0,0]^T\}$

$$D = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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p6 $F(x,y) = \begin{pmatrix} x^2 y \\ e^{x-y} \end{pmatrix}$, $G(u,v) = u + \log v$

$$DF(x,y) = \begin{array}{c|cc} & x & y \\ \hline f_1 & 2xy & x^2 \\ f_2 & e^{x-y} & -e^{x-y} \end{array}$$

$$DF(1,1) = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$D(G \circ F)(1,1) = DG\big|_{F(1,1)} DF(1,1)$$

$$DG = (1, \tfrac{1}{v})$$

$$(1, \tfrac{1}{1}) \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= (3, 0)$$

$$G \circ F(x,y) = x^2 y + x - y$$

$$\gamma(t) = \begin{pmatrix} 1+t \\ 1-t \end{pmatrix}$$

$$\frac{d}{dt} G \circ F(\gamma(t)) =$$

P21

Indicator Function

$$\mathbb{1}_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

$\mathbb{1}_A(x)$ is RV

$$E[\mathbb{1}_A(x)] = \int_{-\infty}^{\infty} \mathbb{1}_A(x) f(x)\,dx$$

$$= \int_A f(x)dx = \mathbb{P}(A)$$

Fundamental bridge

P1 $K \subseteq \mathbb{R}^n$ compact

$F_1 \supseteq F_2 \supseteq F_3 \ldots$ $\forall$ $F_i \neq \emptyset$

& F_i closed

wts $\bigcap_{i=1}^{\infty} F_i \neq \emptyset$

Pick $x_i \in F_i$ $\forall i$,

gives $\{x_i\}_{i=1}^{n} \subseteq K$

since K compact,

$\exists \{x_{i_k}\}_{k=1}^{\infty}$ s.t. $\lim_{k \to \infty} x_{i_k} \to x^* \in K$

$\forall N \in \mathbb{Z}^+$, for all $i_k \geq N$

s.t. $x_{i_k} \in F_{i_k} \subseteq F_N$

since F_N closed, contains limit points

since x_{ik} convergent sbsq.,

$x^* \in F_N$

Since N arbitrary

$x \in \bigcap_{N=1}^{\infty} F_N$

$\square$

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p20 let $X + Y$ have finite second moments

prove $cov(X,Y) = cov(X, \mathbb{E}[Y|X])$

First show $\mathbb{E}[XY] = \mathbb{E}[X\mathbb{E}[Y|X]]$

law of iterated expectations

$\mathbb{E}[XY] = \mathbb{E}[\mathbb{E}[XY]|X]$

$$\mathbb{E}[X\,\mathbb{E}[Y|X]]$$

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$$
$$= \mathbb{E}[XY - X\mathbb{E}Y - Y\mathbb{E}X + \mathbb{E}X\mathbb{E}Y]$$
$$= \mathbb{E}[XY] - \mathbb{E}[X\mathbb{E}Y] - \mathbb{E}[Y\mathbb{E}X] + \mathbb{E}X\mathbb{E}Y$$
$$= \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y - \cancel{\mathbb{E}Y\mathbb{E}X} + \cancel{\mathbb{E}X\mathbb{E}Y}$$
$$(*) \quad = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y$$

$$\mathrm{Cov}(X,\mathbb{E}(Y|X)) = \mathbb{E}[(X-\mathbb{E}(X))(\mathbb{E}(Y|X) - \mathbb{E}(\mathbb{E}(Y|X)))]$$
$$= \mathbb{E}[X\mathbb{E}(Y|X) - X\mathbb{E}(Y) - \mathbb{E}(X)\mathbb{E}(Y|X) + \mathbb{E}(X)\mathbb{E}(Y)]$$
$$= \mathbb{E}[X\mathbb{E}(Y|X)] - \mathbb{E}[X\mathbb{E}(Y)] - \mathbb{E}[\mathbb{E}(X)\mathbb{E}(Y|X)]$$
$$= \mathbb{E}[X\mathbb{E}(Y|X)] - \mathbb{E}(X)\mathbb{E}(Y) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$
same as $(*)$

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p72
let $m(x) = \mathbb{E}[Y|X]$
for any square-integrable function $g(x)$ s.t. $\mathbb{E}[g(x)^2] < \infty$

prove
$$\mathbb{E}[(Y-g(x))^2] = \mathbb{E}[(Y-m(x))^2] + \mathbb{E}[(m(x)-g(x))^2]$$

hint: expand
$$Y - g(x) = (Y-m(x)) + (m(x)-g(x))$$
+ condition on X to eliminate cross term

what does the identity imply about $m(x)$?

$\mathbb{E}[(Y-g(X))^2] =$

$$\mathbb{E}[((Y-m(X))+(m(X)-g(X)))^2]$$

$$\mathbb{E}[(Y-m(X))^2]+2\mathbb{E}[(Y-m(X))(m(X)-g(X))]$$

$$+\mathbb{E}[(m(X)-g(X))^2]$$

Cross term

$$2\Big(\underbrace{\mathbb{E}[(Y(m(X)-g(X))]}_{}-\mathbb{E}[m(X)(m(X)-g(X))]\Big)$$

law of iterated expect $\mathbb{E}[\mathbb{E}[Y(m(X)-g(X))|X]]$

$$= \mathbb{E}[(m(X)-g(X))\,\mathbb{E}[Y|X]]$$

$$= \mathbb{E}[m(X)(m(X)-g(X))] - \mathbb{E}[m(X)(m(X)-g(X))]$$

$$=0$$

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p11 find maximize of

$$f(x,y,z) = -x^2-y^2-z^2+xy+yz+3x-y+4z$$

$$\text{s.t. } x+y\le 2,\quad y+z\ge 4$$

a) show f strictly concave

b) find unconstrained max & check feasibility

c) write out KKT conditions

d) determine active constraints

Step I: rewrite constraints

Step 1: rewrite constraints

$g_1(x,y,z) = 2 - x - y \geq 0 \qquad g_2(x,y,z) = y + z - 4 \geq 0$

a) $\nabla f = \begin{bmatrix} -2x + y + 3 \\ -2y + x + z - 1 \\ -2z + y + 4 \end{bmatrix}$

$$Hf = \begin{array}{c} \\ f_1 \\ f_2 \\ f_3 \end{array}\begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} \end{array}$$

Definiteness: Sylvester's criterion

neg def if $\det|-2|$ neg ✓
-2

$\det\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}$ pos ✓
3

$\det H = -8 + 0 + 0 - 0 + 2 + 2$

-4 neg ✓

$-2(3) - 1(-2)$ cofactor

$-6 + 2 = -4$ same

Hf neg def $\Rightarrow$ f strictly concave

b) $-2x + y + 3 = 0$

$-2y + x + z - 1 = 0$

$-2z + y + 4 = 0$

$$\left[\begin{array}{ccc|c} -2 & 1 & 0 & -3 \\ 1 & -2 & 1 & 1 \\ 0 & 1 & -2 & -4 \end{array}\right]$$

$$-2y + x + z - 1 = 0 \qquad \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 1 & -2 & -4 \end{array}\right]$$

$$-2z + y + 4 = 0$$

$$\xrightarrow{1 \leftrightarrow 2} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ -2 & 1 & 0 & -3 \\ 0 & 1 & -2 & -4 \end{array}\right]$$

$$\xrightarrow{+2R_1} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & -3 & 2 & -1 \\ 0 & 1 & -2 & -4 \end{array}\right]$$

$$\xrightarrow{2 \leftrightarrow 3} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & 2 & -1 \end{array}\right]$$

$$\xrightarrow{+3R_2} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & +4 & +13 \end{array}\right] \qquad -1-12$$

$$z = 13/4$$

$$y = -4 + \frac{26}{4} = \frac{10}{4} = \frac{5}{2}$$

$$-\frac{16}{4}$$

$$x = 1 + 5 - \frac{13}{4} = \frac{11}{4}$$

$$\frac{24}{4}$$

but $x + y > 2$

$\frac{11}{4} + \frac{10}{4}$

$\frac{21}{4} = 5\frac{1}{4} > 2$

unconstrained max not feasible

c) Suppose (x^*, y^*, z^*) local maximizer of constrained problem & CQ holds

then $\exists \lambda^* \in \mathbb{R}^2$ s.t.

(i) stationarity

$$\nabla f(x^*, y^*, z^*) + \sum_{i=1}^{2} \lambda_i^* \nabla g(x^*, y^*, z^*) = 0$$

$$\begin{bmatrix} -2x^* + y^* + 3 \\ -2y^* + x^* + z^* - 1 \\ -2z^* + y^* + 4 \end{bmatrix} + \lambda_1^* \underbrace{\begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}}_{\nabla g_1} + \lambda_2^* \underbrace{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}_{\nabla g_2} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

ii) primal feasibility
constraint satisfaction

$g_1 : 2 - x^* - y^* \geq 0$

$g_2 : y^* + z^* - 4 \geq 0$

$g_2: y^* + z^* - 4 \geq 0$

iii) dual feasibility

$\lambda_1^* \geq 0 \quad \lambda_2^* \geq 0$

iv) complementary slackness

$\lambda_1^*(2 - x^* - y^*) = 0$

$\lambda_2^*(y^* + z^* - 4) = 0$

d) i) if $\lambda_1^* = 0$ & $\lambda_2^* = 0$,

same as unconstrained,

already shown not feasible

ii) if $\lambda_1^* > 0$, $\lambda_2^* > 0$

∇L

$L_x = -2x + y + 3 - \lambda_1 = 0$

$L_y = -2y + x + z - \lambda_1 + \lambda_2 = 0$

$L_z = -2z + y + 4 + \lambda_2 = 0$

$L_{\lambda_1} = 2 - x - y = 0$

$L_{\lambda_2} = y + z - 4 = 0$

$-2x + y + 3 - \lambda_1 = 2 - x - y$

$-x + 2y + 1 = \lambda_1$

$-2z + y + 4 + \lambda_2 = y + z - 4$

$-3z + 8 = -\lambda_2$

$\mathcal{L}_{\lambda_2} = y + z - 4 = 0$

$-3z + 8 = -\lambda_2$

$3z - 8 = \lambda_2$

$-2y + x + z - (-x + 2y + 1) + (3z - 8) = 0$

$2x - 4y + 4z - 9 = 0$

$x = 2 - y$

$z = 4 - y$

$4 - 4y - 4y + 16 - 4y$

$11 = 12y$

$y = 11/12$

$x = \frac{24 - 11}{12} = \frac{13}{12}$

$z = \frac{48 - 11}{12} = \frac{37}{12}$

e) Unique global max

b/c strictly concave obj. funct

& constraint region closed

& convex

& Linear independence constraint qual. holds

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p4 $S = \bigcup_{n=1}^{\infty} \left(\{ \frac{1}{n} \} \times [0, 1] \right) \subseteq \mathbb{R}^2$

a) show S bounded but not closed

$S_i = \{ \frac{1}{i} \} \times [0, 1]$

bounded

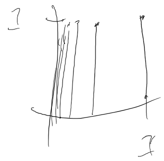
$\exists\, x \in \mathbb{R}^2 .\ r > 0 \le t.$

$-9 = 0$

$y + z = 4$

$\frac{11}{12} + \frac{37}{12}$

$\frac{48}{12} = 4$



bounded

$\exists\, x \in \mathbb{R}^2,\ r > 0$ s.t.

$\forall y \in S,\ \|y - x\| < r$

$\sup \frac{1}{i} = \frac{1}{1} = 1$ & $\inf \frac{1}{i} = \frac{1}{\infty} = 0$

so $S_i \subseteq [0,1] \times [0,1]$

not closed, doesn't contain limit point

Construct sequence $\{x_k\}_{k=1}^{\infty} = \left(\frac{1}{k}, y\right) : 0 \le y \le 1$

$\lim_{k \to \infty} \frac{1}{k} = 0 \Rightarrow \{x_k\}_{k=1}^{\infty} \to (0, y)$ but $(0,y) \notin S$

convergent seq. has limit not in set,

set is not closed

b)

Aug 27

Implicit Function

$$F(x,y,z) = \begin{bmatrix} x + y + z - 3 \\ xy + 2z - 3 \end{bmatrix}$$

$y(x) = \phi(x)$ $\quad y(x) = 3 - x - z \Rightarrow y(1) = ?$

$y(x) = \phi(x)$ $\quad y(x) =$

$z(x) = \psi(x)$ $\quad y(x) =$

1) Check Jacobian $F(1,1,1) = 0$

$$D_{(y,z)} F(1,1,1) = \begin{bmatrix} 1 & 1 \\ x & 2 \end{bmatrix}\Bigg|_{(1,1,1)} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

2) Calculate Derivative Around $(1,1,1)$

$$F_{(y,z)} = \begin{cases} x + \phi(x) + \psi(x) - 3 = 0 \\ x\phi(x) + 2\psi(x) - 3 = 0 \end{cases}$$

Der wrt x

$$D_x F_{(y,z)} = \begin{bmatrix} 1 + \phi'(x) + \psi'(x) = 0 \\ \phi(x) + x\phi'(x) + 2\psi'(x) = 0 \end{bmatrix}$$

$$D_x F_{(y,z)}(1,1,1) = 1 + \phi'(1) + \psi'(1) = 0$$

$$\phi(1) + (1)\phi'(1) + 2\psi'(1) = 0$$

why 1?

$(1,1,1)$ by assumption & $y = 1$ also consistent

by assumption

$\phi'(1) = -1$ $\qquad 1 - 1 + 0 = 0$

$\psi'(1) = 0$ $\qquad 1 + (-1) + 0 = 0$ ✓

from Jacobian

also consistent system means $y(1) = 1$ must hold

3) Check 2nd der wrt x

$$\phi''(x) + \psi''(x) = 0$$

$$\phi'(x) + \phi'(x) + x\phi''(x) + 2\psi''(x) = 0$$

at $x = 1$

$$\phi''(1) + \psi''(1) = 0$$

$$2\phi'(1) + (1)\phi''(1) + 2\psi''(1) = 0$$

$\phi''(1) = -2$ $\qquad -2 + 2 = 0$ ✓

$\psi''(1) = 2$

from Hessian $\qquad 2(-1) + (-2) + 2(2) = 0$ ✓

4) Taylor Approx

b) Taylor Approx

at $x_0 = 1$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + o((x-x_0)^2)$$

$$\phi(x) = \phi(1) + \phi'(1)(x-1) + \frac{1}{2}\phi''(1)(x-1)^2 + o((x-1)^2)$$

$$\psi(x) = \psi(1) + \psi'(1)(x-1) + \frac{1}{2}\psi''(1)(x-1)^2 + o((x-1)^2)$$

$$\Rightarrow \phi(x) = 1 - 1(x-1) - (x-1)^2 + o((x-x_0)^2)$$

$$\psi(x) = 1 + (x-1)^2 + o((x-x_0)^2)$$

p15

$$\max_{x,y} \; xy - y^2 + 2y$$

$$\text{s.t.} \quad 0 \le x \le 2$$

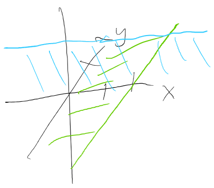
$$0 \le x + y \le 2$$

$$\Updownarrow$$

$$0 \le y \le 2 - x$$

for fixed x:

$$\max_{y} \; xy - y^2 + 2y$$



$$f'(y) = x - 2y + 2 = 0$$

$$2y = x + 2$$

$$\boxed{y^*_{(x)} = \frac{1}{2}(x+2)}$$

$f''(y) = -2$ strictly concave

if $y^* = \frac{1}{2}x + 1 \leq 2 - x$

$\frac{3}{2}x \leq 1$

$x \leq 2/3$

piecewise max

$$f(y^*(x)) = \begin{cases} f(\frac{1}{2}(x+2)) & \text{if } 0 \leq x \leq \frac{2}{3} \\ f(2-x) & \text{if } \frac{2}{3} < x \end{cases}$$

value function $V(x)$

evaluate for x given $f(x)$

then $\max_x V(x) = \max_x f(y^*(x)) = \begin{cases} \frac{1}{4}x^2 + x + 1 & \text{if } 0 \leq x \\ -2x^2 + 4x & \text{other} \end{cases}$

$2\frac{2}{3}$

$$V'(x) = \begin{cases} \frac{1}{2}x + 1 & 0 \le x \le 2/3 \\ -4x + 4 & \text{o.w.} \end{cases}$$

$\frac{1}{2}x + 1 = 0 \Rightarrow x = -2$
infeasible

$-4x + 4 = 0 \Rightarrow \boxed{x^* = 1}$ ✓

so $V''(1) = -4 < 0$
strictly concave

check border $V\left(\frac{2}{3}\right) = \frac{1}{2}\left(\frac{2}{3}\right) + 1 = \frac{4}{3}$

& $V(x^*) = V(1) = -2(1)^2 + 4($

$V(1) > V($

so x^* max m

Find $y^*(x^*)$

$f(y^*(x^*))$ is in $\frac{2}{3} < x \le 2$

since $x^* = 1$

$y^* = \frac{1}{2}(x+2) =$

$\frac{3}{2}$?

$\frac{}{3}$

$1) = 2$

$^{2}/_{3})$

zu

but $0 \leq y \leq 2-x$
so $0 \leq y \leq 1$
$3/2$ unfeasible

$f(y^*(x)) = f(2-x)$ if $\frac{2}{3} < x \leq 2$

$y^*(1) = 2 - x = 1$
so $f(x^*, y^*) = f(1,1)$
$= (1)(1) - (1)^2 + 2(1$
$1 - 1 + 2 =$

p20 Cov

X, Y finite second moment
note: Var isn't Second. 2nd mom
it's centered around mean

Var $E(X - E[X])^2$

2nd mom uncent $E[X]^2$

$$Cov(X,Y) = Cov(X, E[Y|X])$$
$$= E\{X E[Y|X]\} - E[X] E[E[Y|X]]$$

$$= \mathbb{E}\{X\,\mathbb{E}[Y|X]\} - \mathbb{E}[X]\,\mathbb{E}[\mathbb{E}[Y|X]]$$

Lin LIE

$$= \mathbb{E}\{\mathbb{E}[XY|X]\} - \mathbb{E}[X]\,\mathbb{E}[Y]$$

LIE

$$\mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$$

$$\mathrm{cov}(X,Y)$$